

EXTENSION 10I Similar Triangles

Learning intentions

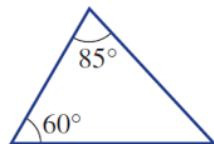
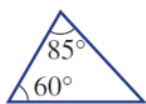
- To understand that determining whether triangles are similar can be done using the similarity tests AAA, SSS, SAS and RHS
- To be able to decide if two triangles are similar
- To be able to find missing lengths using a pair of similar triangles

Key Ideas

Two triangles are similar if:

AAA

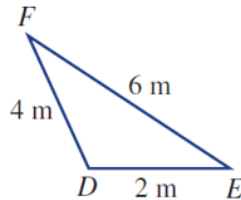
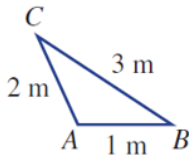
- Three pairs of corresponding angles are equal. If two pairs are known, then the angle sum of a triangle can be used to show that the third pair of angles are also equal.



$$\begin{aligned}\text{Missing angle} &= 180^\circ - (60^\circ + 85^\circ) \\ &= 35^\circ\end{aligned}$$

SSS

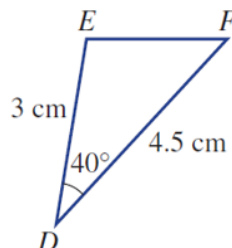
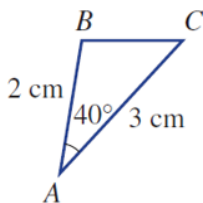
- All pairs of corresponding sides are in the same ratio.



$$\frac{DE}{AB} = \frac{EF}{BC} = \frac{DF}{AC} = 2$$

SAS

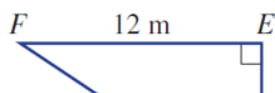
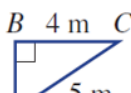
- Two pairs of corresponding sides are in the same ratio and the included angles are equal.



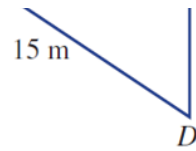
$$\frac{DE}{AB} = \frac{DF}{AC} = 1.5$$

RHS

- A pair of right angles, the pair of hypotenuse lengths and another pair of corresponding sides are in the same ratio.



$$\frac{DF}{AC} = \frac{FE}{BC} = 3$$

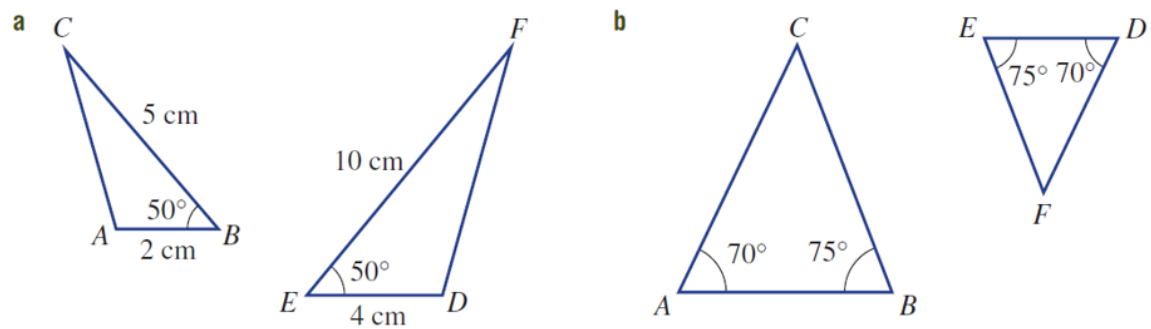


■ If two triangles $\triangle ABC$ and $\triangle DEF$ are similar, we write $\triangle ABC \sim \triangle DEF$ or $\triangle ABC \parallel \triangle DEF$.

- This is called a similarity statement.
- Letters are usually written in matching order.

Example 15 | Explaining why two triangles are similar

Explain, with reasons, why these pairs of triangles are similar.



✗

